

SymPy Bug Card

Bug 41: Unsound Extended-Real Assumption for a Reciprocal

Task. Decide whether the reciprocal is extended real for an arbitrary real symbol xr .

$$\text{ask}\left(Q.\text{extended_real}\left(\frac{1}{xr-1}\right)\right)$$

SymPy's answer.

$$\text{ask}\left(Q.\text{extended_real}\left(\frac{1}{xr-1}\right)\right) = \text{True}.$$

Correct answer. SymPy should not return True; the answer should be False or unknown without a nonzero-denominator assumption.

Explanation. The real substitution $xr = 1$ is allowed, but it gives

$$\frac{1}{1-1} = \text{zoo},$$

and zoo is not in the extended real line $\mathbb{R} \cup \{-\infty, +\infty\}$. Thus the universal predicate conclusion is contradicted by a valid real specialization.

Likely root cause. The `ExtendedRealPredicate` handler in `sympy/assumptions/handlers/sets.py` applies closed-group reasoning to `Pow`, so $(xr-1)^{-1}$ is accepted because both its base and exponent are extended real; it does not require the reciprocal base to be provably nonzero. This belongs to a known reciprocal-with-possibly-zero-denominator unsoundness family in the assumptions handlers, publicly tracked as issue [#28150](#) (open) with PRs [#28164](#) (closed) and [#28377](#) (open). Those address `Q.real` on a concrete-number base `Pow(0, -1)`; this `Q.extended_real` symbolic-base manifestation is a distinct, still-unfixed instance that the referenced fixes would not resolve.

Artifact (GitHub).

[bug-report/bug-041-unsound-extended-real-assumption-for-a-reciprocal/](#)